Practice Set 6 Paper 2 – Urology

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Practice paper compiled from the question bank.

Q1. Discuss treatment of CRPC

Type: LAQ (25 marks) (also answered in 2020-Summer Paper 2, Q3a)

Definition (EAU [European Association of Urology])

- CRPC (castration-resistant prostate cancer): castrate testosterone **<50 ng/dL (1.7 nmol/L)** plus either:
  - **Biochemical progression:** three consecutive PSA (prostate-specific antigen) rises 1 week apart, two of them 50% above nadir, PSA >2 ng/mL; or
  - **Radiological progression:** ≥2 new bone lesions, or soft-tissue progression by RECIST (Response Evaluation Criteria in Solid Tumours).

Why it happens: the AR still drives growth

- Castration-resistant is **not** androgen-independent, so treatment first targets the AR (androgen receptor) more deeply.
- Mechanisms: AR overexpression and amplification; AR mutation (activated by other steroids); **splice variants lacking the ligand-binding domain**; adrenal and intratumoural androgen synthesis (CYP17 overexpressed); bypass pathways (PI3K/AKT, PTEN loss in ~40%).
- **Lineage plasticity:** neuroendocrine change with low PSA, visceral and lytic disease.

Evaluation

- Testosterone (exclude non-compliance), PSA, **PSADT (PSA doubling time)**; CT and bone scan; PSMA (prostate-specific membrane antigen) PET.
- Prognosis: performance status, pain, haemoglobin, ALP (alkaline phosphatase), LDH (lactate dehydrogenase), visceral disease.
- **Germline testing** in all metastatic disease; somatic sequencing of tumour tissue for HRR (homologous recombination repair) genes and MSI (microsatellite instability).
- Biopsy a metastasis if progression is fast with a low PSA.
- PCWG3 (Prostate Cancer Clinical Trials Working Group 3): early bone-scan lesions may be **flare**; re-baseline at 12–16 weeks, then progression = ≥2 new lesions.

Clinical states model (PCWG3):

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Localised → rising PSA → mCSPC (castration-sensitive)
                  ↓ (on ADT)
              nmCRPC → mCRPC 1st line → 2nd line → 3rd line
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General principles

1. **Continue ADT (androgen deprivation therapy) lifelong**, including during chemotherapy.
2. **Antiandrogen withdrawal** (e.g. bicalutamide): watch PSA 4–8 weeks; brief responses in a minority.
3. Choose by prior therapy, symptoms, visceral disease, genomics and comorbidity.
4. **Switch mechanism.** Cross-resistance means a second ARPI (AR-pathway inhibitor) helps only 15–30%.

nmCRPC (non-metastatic)

- **PSADT ≤10 months:** add an ARPI. Each improved MFS (metastasis-free survival) and overall survival:

    - Apalutamide 240 mg (**SPARTAN**: MFS HR 0.28); enzalutamide 160 mg (**PROSPER**); darolutamide 300 mg twice daily (**ARAMIS**: MFS 40.4 vs 18.4 months).
- Slower PSADT: continue ADT and monitor.

mCRPC: life-prolonging options

Class	Agent and dose	Key trial (overall survival)
ARPI	**Abiraterone** 1000 mg fasting + prednisone	COU-AA-301: 14.8 vs 10.9 months; COU-AA-302 (pre-chemotherapy)
	Enzalutamide 160 mg	AFFIRM: 18.4 vs 13.6 months; PREVAIL
Taxane	**Docetaxel** 75 mg/m² 3-weekly + prednisone	TAX 327: 18.9 vs 16.4 months (mitoxantrone)
	Cabazitaxel 20–25 mg/m² after docetaxel	TROPIC: 15.1 vs 12.7; CARD: beats a second ARPI
Radioligand	**Radium-223** 50 kBq/kg 4-weekly ×6 (symptomatic bone-only, no visceral)	ALSYMPCA: 14.9 vs 11.3 months
	^{177}Lu-PSMA-617 (PSMA-positive, after ARPI + taxane)	VISION: 15.3 vs 11.3 months
PARP (poly-ADP-ribose polymerase) inhibitor	Olaparib (HRR genes); rucaparib (BRCA)	PROfound; TRITON3
	Olaparib + abiraterone; niraparib + abiraterone; talazoparib + enzalutamide (first line)	PROpel; MAGNITUDE; TALAPRO-2
Immunotherapy	**Sipuleucel-T** (asymptomatic, no visceral disease)	IMPACT: 25.8 vs 21.7 months
	Pembrolizumab for MSI-high/dMMR (mismatch repair-deficient), 2–5%	Tumour-agnostic approval

- **Abiraterone** blocks CYP17, which also lowers cortisol. ACTH (adrenocorticotropic hormone) then rises and causes mineralocorticoid excess (hypokalaemia, hypertension, oedema), so prednisone is co-prescribed. Never combine with radium-223 (more deaths and fractures).
- **Enzalutamide:** seizures (~1%), fatigue, falls. Apalutamide: rash, hypothyroidism.
- **PARP inhibitors** cause synthetic lethality in HRR-deficient cells; benefit is greatest with BRCA2 and weak with ATM.

Sequencing by prior mCSPC therapy

Prior therapy	Next option
Docetaxel only	ARPI
ARPI only	Docetaxel, radium-223, sipuleucel-T
Both	Cabazitaxel, ^{177}Lu-PSMA, PARP inhibitor if HRR mutation

Neuroendocrine/aggressive variant

- Low PSA, liver or brain metastases, lytic bone lesions with hypercalcaemia. Treat with **platinum + etoposide** (or carboplatin + taxane) ± radiotherapy; survival <12 months.

Supportive and palliative care

- **Bone health:** denosumab 120 mg SC or zoledronic acid 4 mg IV (renal dose adjustment) every 4 weeks, with calcium and vitamin D. Denosumab delayed the first SRE (skeletal-

related event) longer (20.7 vs 17.1 months); neither prolongs life. Risk: osteonecrosis of the jaw.

- Focal bone pain: external-beam radiotherapy.
- **Cord compression:** high-dose dexamethasone → urgent MRI → radiotherapy ± decompression (Patchell trial: surgery + radiotherapy better).
- Ureteric obstruction: stent or nephrostomy. Anaemia; hospice care.

Recent advances

- PSMA-ligand therapy earlier (PSMAfore); AR degraders (ARV-110); AKT inhibition in PTEN loss; bipolar androgen therapy; antibody–drug conjugates.

Diagram to draw: clinical states model with drugs placed at each state.

> **Key points to remember:**
>
> - Testosterone <50 ng/dL with PSA or radiological progression; keep ADT going.
> - nmCRPC with PSADT ≤10 months: apalutamide, enzalutamide or darolutamide.
> - mCRPC: ARPI, docetaxel, cabazitaxel, radium-223, ^{177}Lu-PSMA, PARP inhibitors, sipuleucel-T.
> - Switch mechanism; never radium-223 with abiraterone.

Q2. Classification and treatment of lower track trauma in # pelvis

Type: LAQ (25 marks)

Assumed meaning: lower urinary tract trauma (bladder and urethra) in pelvic fracture.

Introduction

- Pelvic fracture accompanies **83–95% of bladder injuries**, yet only **5–10% of pelvic fractures** injure the bladder. Urethral injury occurs in about **10% of male** and up to 6% of female pelvic fractures.
- 80–94% have other major injuries, which cause the deaths; long-term disability comes from wrong management.

Mechanism

- **EPR (extraperitoneal rupture):** pelvic disruption tears the bladder at its fascial attachments, or bone spicules lacerate it.
- **IPR (intraperitoneal rupture):** a blow on a **full bladder** bursts the dome, the weakest part covered by peritoneum.
- **PFUI (pelvic fracture urethral injury):** the posterior urethra is fixed by the puboprostatic ligaments, so displacement tears it at the **bulbomembranous junction**. The sphincter is usually avulsed but stays functional. Highest risk: straddle fracture (all four rami), vertically unstable fractures.

Classification

Bladder: AAST (American Association for the Surgery of Trauma)

Grade	Injury
I	Contusion, intramural haematoma, partial-thickness laceration
II	Extraperitoneal wall laceration <2 cm
III	Extraperitoneal ≥2 cm, or intraperitoneal <2 cm
IV	Intraperitoneal ≥2 cm
V	Laceration extending into bladder neck or ureteric orifice (trigone)

- Practical: contusion, EPR (simple or complex), IPR, combined.

Urethra: Goldman

Type	Injury
I	Posterior urethra stretched, intact
II	Disruption above the urogenital diaphragm
III	Disruption of membranous urethra through the urogenital diaphragm (commonest)
IV	Bladder neck injury extending into urethra (IVa: bladder-base injury mimicking IV)
V	Anterior urethral injury

- Chapter's practical grading: **partial injury, minor distraction, major distraction** (a cystoscope cannot pass into the bladder). Stricture risk rises with each.

Clinical features

- Bladder: **gross haematuria (nearly all)**, suprapubic pain, inability to void, free fluid.
- Urethra: **triad: blood at the meatus, inability to void, palpable bladder**. A "high-riding" prostate and butterfly perineal haematoma are unreliable. **Vaginal examination** in every female.

Investigations

- **RUG (retrograde urethrogram) first if blood at the meatus** (AUA [American Urological Association]). Method: 16 Fr catheter in fossa navicularis, 3 mL balloon, penile traction, oblique position, 20 mL contrast, film during injection. Females: urethroscopy.
- **Cystography:** absolute indication: **gross haematuria + pelvic fracture** (rupture in ~29%). Relative: gross haematuria alone; microscopic haematuria with fracture.
  - **CT cystography:** retrograde fill with 350 mL of 2–4% contrast; no drainage film needed.
  - Plain: scout, full-bladder and **drainage films** (posterior leaks otherwise missed).
  - Clamping the catheter after IV contrast is inadequate.
- **EPR:** flame-shaped pelvic contrast. **IPR:** contrast outlines bowel loops.

Treatment

Resuscitation first (ATLS [Advanced Trauma Life Support]); pelvic stabilisation.

Bladder

- **Uncomplicated EPR:** 22 Fr urethral catheter 2–3 weeks; cystogram before removal; repair if leak persists beyond 4 weeks.
- **Immediate repair:** IPR; penetrating injury; bladder neck injury; bone fragments in the bladder; rectal or vaginal injury; poor drainage or clots; laparotomy for other reasons.
- IPR: open or laparoscopic two-layer closure; inspect ureteric orifices; cystogram at 7–10 days. At laparotomy, repair EPR from inside the bladder, leaving the pelvic haematoma undisturbed.
- Suprapubic tube adds nothing over urethral catheter (AUA). Antibiotics only around IPR repair.
- Rectal or vaginal injury: separate the walls, no overlapping suture lines, interpose tissue.

Urethra: acute

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Blood at meatus → RUG
  Partial → one gentle catheter attempt (confirm position)
  Complete → SPC (suprapubic catheter): percutaneous 16 Fr
             or open (if bladder repair needed)
             ± brief primary realignment if stable
             (flexible cystoscopy, antegrade + retrograde)
  Bladder neck/rectal injury → immediate repair
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- Avoid small "punch" tubes. SPC is safe with pelvic plating (AUA); place it high.
- Realignment: catheter out at 4–6 weeks, keep SPC. It does **not** lower urethroplasty rates; most still form a 1–2 cm stenosis.

Urethra: delayed (2–3 months)

- **Up-and-down-o-gram** (cystogram + RUG) for gap length; MRI for complex cases; flexible cystoscopy if bladder neck does not open.
- **Perineal excision and primary anastomosis (EPA)** is the treatment of choice: lithotomy → mobilise bulbar urethra → excise all scar → identify prostatic apex → tension-free spatulated anastomosis.
- Escalating steps for long gaps: **corporal splitting → inferior pubectomy → supracrural rerouting**. Abdominoperineal approach for fistula, bladder neck injury, failed repair, children.
- **Not** "cut-to-the-light" or repeated DVIU (direct visual internal urethrotomy) for gaps ≥1 cm.

Complications

- ED (erectile dysfunction) up to 82%, mainly from the injury (nerve, arterial, venous).

- Restenosis 5–15% (DVIU usually works); incontinence <4%.
- Missed IPR: ascites, acidosis, azotaemia (peritoneal reabsorption).
- Fistula, abscess, neurogenic voiding dysfunction; follow up ≥1 year.

Diagram to draw: pelvis with EPR (flame-shaped) vs IPR, and bulbomembranous distraction defect.

> **Key points to remember:**
> - Gross haematuria + pelvic fracture = retrograde CT cystogram (350 mL).
> - EPR: catheter 2–3 weeks; IPR: immediate repair.
> - PFUI: SPC, then perineal anastomotic urethroplasty at 2–3 months.
> - Bladder neck or rectal injury: repair at once.

Q3a. Pre-emptive renal transplant

Type: SAQ (10 marks)

Definition

- Kidney transplantation **before starting maintenance dialysis** in a patient with advanced CKD (chronic kidney disease).

Rationale

- Every year on dialysis adds cardiovascular disease, vascular calcification and bone–mineral damage that the graft later inherits. Pre-emptive transplantation **avoids this exposure**.
- Transplantation beats dialysis: Wolfe (1999) showed early risk is higher, equal at 106 days, then 68% lower long term.

Advantages

- Better patient and graft survival than transplantation after dialysis; longer dialysis time worsens outcome (registry data).
- No dialysis access surgery, access infection or catheter complications.
- Fewer transfusions, so less HLA (human leucocyte antigen) sensitisation.
- Better quality of life, employment; lower cost.
- Children: better growth and development.

Frequency

- 28.7% of US transplants in 2021 (living donor 51%, deceased donor 21%, children 56%). Commonest with a **living donor**.

Timing and evaluation

- Evaluate early, ideally before dialysis. KDIGO (Kidney Disease: Improving Global Outcomes) 2020: refer when eGFR (estimated glomerular filtration rate) is <30 mL/min/1.73 m^2, 6–12 months before expected dialysis. Living-donor transplant is usually done at eGFR <20 with progressive decline.
- Deceased-donor listing (US) at eGFR ≤20.
- Full recipient work-up: cardiac, vascular (iliac calcification), infection (TB [tuberculosis], hepatitis B and C, HIV [human immunodeficiency virus]), malignancy, psychosocial; recurrence risk (primary FSGS [focal segmental glomerulosclerosis]).
- **Urological:** bladder function and lower tract (valves, neurogenic bladder); native nephrectomy only for specific indications.
- India: mainly living related donors under THOA (Transplantation of Human Organs Act).

Limitations

- Needs early referral and a ready donor.
- Transplant may come earlier than needed, using up dialysis-free native function; adherence untested.
- In advanced CKD, bowel augmentation may speed decline.

Also know: ABO-incompatible or paired exchange programmes widen pre-emptive options.

Diagram to draw: Wolfe curve: relative risk of death after transplant vs dialysis over time.

> **Key points to remember:**
>
> - Transplant before dialysis = best outcome.
> - Refer at eGFR <30; transplant usually at eGFR <20.
> - Commonest with living donors and children.

Q3b. What are the causes of bleeding in PCNL? How will you manage persistent hematuria 6 days after PCNL!?

Type: SAQ (10 marks) (PCNL complications also in Practice Set 2 Paper 2, Q3e)

Introduction

- Bleeding is the commonest serious complication of PCNL (percutaneous nephrolithotomy). Transfusion was historically ~7%, now nearer 1% with smaller tracts. Angioembolisation is needed in <1%.

Causes

Timing	Cause
Intraoperative	Parenchymal or venous tract bleeding; infundibular artery injury; through-and-through puncture; posterior segmental artery injury (upper-pole access); torn infundibulum or pelvis
Early postoperative	Tract bleeding after sheath removal; clot retention
Delayed (about 1 week)	**Pseudoaneurysm**, **AV (arteriovenous) fistula**

- **Risk factors:** infundibular or upper-pole puncture, multiple or large tracts, staghorn stone, long operation, antithrombotic drugs, coagulopathy, diabetes.
- **Prevention:** puncture through the **calyceal fornix** along a posterior calyx (Sampaio: no arterial injury); correct clotting; mini-tracts; tranexamic acid reduces transfusion (Cochrane).

Mechanism of delayed bleeding

- **Pseudoaneurysm:** clot seals an arterial tear, then ruptures intermittently.
- **AV fistula:** artery and vein heal together; the thin vein cannot withstand arterial pressure.

Management at day 6

- At day 6, assume an **arterial lesion** until proved otherwise.

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Resuscitate: IV access, crossmatch, transfuse; Hb, clotting
→ Stop anticoagulants; catheter, irrigate clots
→ Stable, mild: bed rest, clamp nephrostomy, serial Hb
→ Persistent or recurrent bleeding or Hb fall
   → CT angiography (arterial blush) or direct angiography
→ SUPERSELECTIVE ANGIOEMBOLISATION (coils)
→ Fails or unstable after embolisation
   → open exploration, partial or total nephrectomy
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- **Angiography** diagnoses >90% and treats in the same sitting. It may miss intermittent bleeds; non-selective embolisation loses renal tissue.
- Clot colic or obstruction: double-J stent or nephrostomy.
- Follow-up: Hb and creatinine; check for infection and abscess.

Intraoperative ladder (for completeness)

- Pressure on tract → nephrostomy clamped (tamponade) → balloon reinflation or **Kaye tamponade catheter** → angioembolisation.

Diagram to draw: angiogram sketch: pseudoaneurysm before and after coiling.

Key points to remember:

- Early bleeding venous: tamponade.
- Bleeding at about 1 week arterial: pseudoaneurysm or AV fistula.
- Treatment: superselective angioembolisation.

Q3c. Partial nephrectomy – indications and approaches.

Type: SAQ (10 marks) (nephron-sparing methods also in 2022-Summer Paper 3, Q1)

Definition

- PN (partial nephrectomy): complete removal of a tumour with maximal preservation of functioning parenchyma. Equal cancer control to RN (radical nephrectomy) with less CKD (chronic kidney disease) and cardiovascular risk.

Indications

Category	Situations
Imperative	Solitary kidney; bilateral tumours; RN would cause dialysis
Relative	Contralateral kidney threatened (stones, renal artery stenosis, diabetes, hypertension); hereditary RCC (renal cell carcinoma)
Elective	**cT1a with normal contralateral kidney: standard of care**; cT1b–T2 when feasible
Benign	Symptomatic AML (angiomyolipoma); focal atrophy in duplex kidney; infected calyceal diverticulum

- **Relative contraindications:** <20% nephron mass left; pedicle encasement; venous thrombus; cT4; nodes.
- Planning: **R.E.N.A.L. nephrometry score** (Radius, Exophytic, Nearness to collecting system, Anterior/posterior, Location) predicts complexity; 3D models help.

Approaches

Approach	Features
Open, extraperitoneal flank (supracostal 11th/12th rib, 8 cm mini-flank)	Posterior exposure; solitary kidney, complex or central tumours, cold ischaemia
Open transperitoneal (subcostal, chevron, midline)	Bilateral or large tumours, vascular reconstruction
Laparoscopic transperitoneal	Large space, easier suturing; anterior tumours
Laparoscopic retroperitoneal	No bowel mobilisation, direct hilum; posterior tumours
Robot-assisted (RAPN)	Easier suturing, shorter warm ischaemia, fewer conversions

Key technical steps

1. Early hilar control; **intraoperative ultrasound** for borders and depth.
2. Vascular control: artery ± vein clamp; off-clamp or selective clamping for small tumours; ice slush if cold ischaemia needed.
3. Excision: rim of normal tissue, or **enucleation** (hereditary, multifocal). Margin width is unimportant if negative.
4. Renorrhaphy: inner layer (vessels, collecting system) + capsular layer (sliding-clip); drain.

- **Trifecta:** negative margin, no major complication, warm ischaemia ≤25 minutes.
- Function depends mainly on **parenchyma preserved**, less on clamp time.

Complications

- Urine leak (drain → stent); bleeding, **pseudoaneurysm or AV (arteriovenous) fistula** (angioembolisation); AKI (acute kidney injury); positive margin; local recurrence 1–2% elective, 3–5% imperative.

Diagram to draw: tumour excised with rim and two-layer renorrhaphy.

Key points to remember:

- Imperative, relative, elective; PN standard for cT1a.
- Retroperitoneal for posterior, transperitoneal for anterior tumours.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.

Q3d. Adrenal Sx.

Type: SAQ (10 marks) (preoperative phaeochromocytoma management also in 2020-Summer Paper 2, Q3c)

Assumed meaning: surgery of the adrenal gland (adrenalectomy).

Surgical anatomy

- Right: pyramidal; **short vein into posterior IVC (inferior vena cava)**, so it is the riskier side. Left: crescentic; long vein into left renal vein. Three arteries (inferior phrenic, aorta, renal), one vein.

Indications (AAES [American Association of Endocrine Surgeons] 2022)

- **Functional, any size:** aldosteronoma (Conn; after AVS [adrenal venous sampling]), cortisol-secreting adenoma, phaeochromocytoma, sex-hormone tumour, failed pituitary surgery for Cushing disease.
- **Malignant potential:** ≥4 cm; growth; >10 HU (Hounsfield units) unenhanced; washout <50%; suspected ACC (adrenocortical carcinoma); solitary metastasis.
- Myelolipoma >7 cm or symptomatic; cysts.
- **Contraindications (correct first):** uncontrolled hypertension, coagulopathy, peritonitis (transperitoneal); pregnancy relative.

Preoperative preparation

- Biochemistry: metanephrines in **every** case before surgery.
- **Phaeochromocytoma:** α-blockade 2 weeks (phenoxybenzamine or doxazosin), β-blocker only after α, salt and fluids.
- Cushing: hydrocortisone 100 mg IV on the day of surgery. Conn: correct potassium.

Approaches

Approach	Use
Laparoscopic transperitoneal (gold standard since 1992)	Familiar view, larger tumours
Retroperitoneoscopic (lateral or prone)	Prior abdominal surgery, small tumours
Robotic	Either route
Open (subcostal, chevron, thoracoabdominal, flank)	**ACC**, large or invasive tumours

Steps (laparoscopic transperitoneal)

- **Right:** elevate liver → Kocher manoeuvre → follow IVC → **control adrenal vein early** (warn anaesthetist) → free gland to psoas → bag intact.
- **Left:** reflect colon and spleen → follow left renal vein to adrenal vein → clip → free from pancreas and splenic vessels.
- Kidney attachment last (handle). ACC: open en bloc, no rupture.
- **Partial adrenalectomy:** bilateral or hereditary phaeochromocytoma, solitary gland.

Complications

- Bleeding (adrenal vein avulsion, IVC); spleen, pancreatic tail, liver, bowel injury; pneumothorax.
- **Phaeochromocytoma:** surges during handling, **hypotension after vein ligation**, rebound hypoglycaemia.
- **Adrenal insufficiency** after Cushing surgery (~99.7%): steroid taper.
- Conn: **hyperkalaemia** for weeks (high-salt diet).

Diagram to draw: right and left adrenal veins and relations.

Key points to remember:

- Function (any size) or malignancy risk (≥4 cm, >10 HU) indicates surgery.
- Control the vein early; right vein short into IVC.
- Open en bloc surgery for ACC.

Q3e. Malakoplakia

Type: SAQ (10 marks)

Definition

- "Soft plaque": a rare chronic granulomatous inflammation caused by **defective macrophage killing of bacteria**, mostly *E. coli*. Described by Michaelis and Gutmann (1902) and von Hansemann (1903).
- Commonest in the **bladder**; also ureter, kidney, prostate, testis, GI (gastrointestinal) tract, skin, lung.

Pathogenesis

- Macrophages ingest bacteria but cannot digest them (lysosomal defect, low intracellular cGMP [cyclic guanosine monophosphate]).
- Bacterial remnants become mineralised with calcium phosphate, forming laminated **Michaelis–Gutmann bodies**.
- Hosts are often immunocompromised, diabetic, chronically ill or **renal transplant recipients**.

Pathology

- Gross: soft yellow-brown plaques or nodules; renal tumour-like masses.
- Micro: sheets of large foamy histiocytes (**von Hansemann cells**) with **Michaelis–Gutmann bodies** (pathognomonic; "owl's eye" targetoid; stain with von Kossa, PAS [periodic acid–Schiff] and Prussian blue).
- MG bodies may be absent early; **α1-antitrypsin** immunostaining helps.

Clinical features

- Mostly >50 years; urinary tract **female:male 4:1**.
- Bladder: irritative LUTS (lower urinary tract symptoms), haematuria, recurrent *E. coli* UTI (urinary tract infection); plaques or masses mimicking cancer; distal ureteric stricture.
- Kidney: masses, fever, flank pain; renal vein or IVC thrombosis. Testis: epididymo-orchitis. Prostate: mimics carcinoma.

Investigations

- Urine culture (*E. coli*). Cystoscopy and **biopsy is diagnostic**.
- Imaging: enlarged kidneys with filling defects, **no stones, calcification or hydronephrosis** (unlike XGP).

	XGP (xanthogranulomatous pyelonephritis)	Malakoplakia
Cell	Lipid-laden foamy macrophages	Von Hansemann cells with MG bodies
Stone/obstruction	Usually (staghorn)	Absent
Laterality	Unilateral	Uni- or multifocal, bilateral
Treatment	Nephrectomy	Antibiotics; nephrectomy if unilateral, symptomatic

Management

- **Long-term antibiotics active inside macrophages:** fluoroquinolones (concentrate in macrophages), trimethoprim–sulfamethoxazole, rifampicin, doxycycline.
- **Bethanechol + ascorbic acid** raise intracellular cGMP and correct the degradative defect.

- Reduce immunosuppression where possible.
- Surgery if progression despite medical therapy: TUR (transurethral resection) of bladder lesions; **nephrectomy** for symptomatic unilateral renal disease.

Prognosis

- Bilateral or transplant-kidney disease: death often within 6 months; overall mortality can exceed 50%. Unilateral: long survival after nephrectomy.

Diagram to draw: histiocyte with Michaelis–Gutmann bodies.

> **Key points to remember:**
> - Defective macrophage digestion of *E. coli*.
> - Michaelis–Gutmann bodies are pathognomonic.
> - Fluoroquinolones + bethanechol/vitamin C; nephrectomy for unilateral renal disease.

Q3f. Medical treatment of BPH

Type: SAQ (10 marks) (also answered in 2015-Summer Paper 2, Q1)

Rationale

- BPH (benign prostatic hyperplasia) obstruction has a **dynamic** part (α1A-mediated smooth-muscle tone) and a **static** part (DHT [dihydrotestosterone]-driven gland bulk). Drugs target each.

Before treatment

- IPSS (International Prostate Symptom Score), DRE (digital rectal examination), urinalysis, PSA (prostate-specific antigen), PVR (post-void residual); PVR >300 mL = chronic retention.
- **Watchful waiting** with lifestyle advice for mild or non-bothersome LUTS (lower urinary tract symptoms).

Drugs

Class	Doses	Key facts
α1-blockers (first line)	Tamsulosin 0.4 mg; alfuzosin 10 mg; silodosin 8 mg; doxazosin 4–8 mg (titrate)	Act in days; do not prevent AUR (acute urinary retention) or surgery. Hypotension, ejaculatory dysfunction (silodosin most), **IFIS (intraoperative floppy iris syndrome)**
5-ARIs (5α-reductase inhibitors)	Finasteride 5 mg; dutasteride 0.5 mg	Prostate >30–40 mL. Shrink gland ~20%, halve PSA, act after 6 months. PLESS: AUR or surgery ↓51%. Low libido, ED (erectile dysfunction). Also BPH haematuria
Tadalafil 5 mg daily	PDE5 (phosphodiesterase-5) inhibitor	LUTS ± ED; not with nitrates
Antimuscarinic or β3-agonist (mirabegron, vibegron)	Add-on	Persisting storage symptoms; PVR <150 mL
Desmopressin	—	Nocturnal polyuria; monitor sodium

Combination therapy

- **MTOPS (Medical Therapy of Prostatic Symptoms):** progression ↓39% doxazosin, 34% finasteride, **66% combination**. Only finasteride-containing arms reduced AUR and surgery.
- **CombAT (Combination of Avodart and Tamsulosin):** combination reduced AUR or surgery by ~66% vs tamsulosin.
- Combine when **PV (prostate volume) >30–40 mL and PSA >1.5 ng/mL**; α-blocker may stop after 6 months (SMART).
- Tadalafil + finasteride fixed dose approved (FDA [Food and Drug Administration] 2021).

Important cautions

- **IFIS:** billowing iris, prolapse, miosis; stopping the drug does not prevent it.
- 5-ARIs: double the PSA; not for cancer prevention (more high-grade tumours in PCPT [Prostate Cancer Prevention Trial] and REDUCE).
- **AUR:** catheter, α-blocker, then TWOC (trial without catheter).
- **Follow-up:** 4–12 weeks (6 months for 5-ARIs), then yearly; surgery if refractory.

Also know: phytotherapy (saw palmetto) not recommended.

Diagram to draw: algorithm: LUTS → size, PSA, storage symptoms → drug choice.

> **Key points to remember:**
> - α-blockers: rapid relief, no prevention.
> - 5-ARIs: prevent AUR and surgery in glands >30–40 mL.
> - MTOPS combination: progression ↓66%.

Rapid Revision Sheet – Practice Set 6 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss treatment of CRPC (LAQ)

- Testosterone <50 ng/dL with PSA or radiological progression; keep ADT going.
- nmCRPC with PSADT ≤10 months: apalutamide, enzalutamide or darolutamide.
- mCRPC: ARPI, docetaxel, cabazitaxel, radium-223, ^{177}Lu-PSMA, PARP inhibitors, sipuleucel-T.
- Switch mechanism; never radium-223 with abiraterone.

Q2. Classification and treatment of lower track trauma in # pelvis (LAQ)

- Gross haematuria + pelvic fracture = retrograde CT cystogram (350 mL).
- EPR: catheter 2–3 weeks; IPR: immediate repair.
- PFUI: SPC, then perineal anastomotic urethroplasty at 2–3 months.
- Bladder neck or rectal injury: repair at once.

Q3a. Pre-emptive renal transplant (SAQ)

- Transplant before dialysis = best outcome.
- Refer at eGFR <30; transplant usually at eGFR <20.
- Commonest with living donors and children.

Q3b. What are the causes of bleeding in PCNL? How will you manage persistent hematuria 6 days after PCNL!? (SAQ)

- Early bleeding venous: tamponade.
- Bleeding at about 1 week arterial: pseudoaneurysm or AV fistula.
- Treatment: superselective angioembolisation.

Q3c. Partial nephrectomy – indications and approaches. (SAQ)

- Imperative, relative, elective; PN standard for cT1a.
- Retroperitoneal for posterior, transperitoneal for anterior tumours.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.

Q3d. Adrenal Sx. (SAQ)

- Function (any size) or malignancy risk (≥4 cm, >10 HU) indicates surgery.
- Control the vein early; right vein short into IVC.
- Open en bloc surgery for ACC.

Q3e. Malakoplakia (SAQ)

- Defective macrophage digestion of *E. coli*.
- Michaelis–Gutmann bodies are pathognomonic.
- Fluoroquinolones + bethanechol/vitamin C; nephrectomy for unilateral renal disease.

Q3f. Medical treatment of BPH (SAQ)

- α-blockers: rapid relief, no prevention.
- 5-ARIs: prevent AUR and surgery in glands >30–40 mL.
- MTOPS combination: progression ↓66%.